MATH 345 Skeleton Notes
Unit 5: Optimization and training
Section 5.6: Least squares from gradients

The least squares problem

We now revisit least squares as an optimization problem. In Unit 4, residual orthogonality led to the normal equations. Here we obtain the same equations by minimizing a loss function and setting its gradient equal to zero.

For a matrix A, consider the least-squares problem

$$\min_{\mathbf{x} \in \mathbb{R}^n} \|A\mathbf{x} - \mathbf{b}\|^2.$$

This asks for the minimum of the quantity $\|A\mathbf{x} - \mathbf{b}\|^2$. We now use the methods of optimization we have developed to solve this problem in a series of exercises.

Activity

Compute the gradient and Jacobians of the following functions:

Task The gradient of $g(\mathbf{x}) = \mathbf{d} \cdot \mathbf{x}$, for a fixed vector $\mathbf{d}$.

Task The gradient of $f(\mathbf{x}) = \mathbf{x}^T A\mathbf{x}$, for an $n \times n$ matrix A.

Task The Jacobian of the vector-valued function $\mathbf{F}(\mathbf{x}) = C\mathbf{x}$, where C is a fixed $m \times n$ matrix.

Activity

Let A be an $m \times n$ matrix with linearly independent columns and suppose $\mathbf{b} \in \mathbb{R}^m$. Consider the least squares problem

$$\min_{\mathbf{x} \in \mathbb{R}^n} \|A\mathbf{x} - \mathbf{b}\|^2.$$

Task Find the critical points of the function $h(\mathbf{x}) = \|A\mathbf{x} - \mathbf{b}\|^2$.

Activity: Same normal equations in code

Consider the code

```
xhat = np.linalg.lstsq(A, b, rcond=None)[0]
r = b - A @ xhat
A.T @ r
```

1. What is `xhat`?
2. What is `r`?
3. What condition is checked by `A.T @ r`?

4. How does this connect the Unit 4 and Unit 5 viewpoints?

Tags. `[U5-L06, U4-L04 | C+T | Core]`

Activity

Determine whether any critical point is a local minimum, local maximum, or saddle point. Using this information, solve the least squares optimization problem.

This equation is not new. In Unit 4, the normal equations came from residual orthogonality. Now we derive the same equations by minimizing $h(\mathbf{x}) = \|A\mathbf{x} - \mathbf{b}\|^2$. The condition $\nabla h(\mathbf{x}) = \mathbf{0}$ gives $A^T A\mathbf{x} = A^T\mathbf{b}$.

- **Viewpoint:** Projection
  Key condition: Residual orthogonal to $\operatorname{col}(A)$
  Equation: $A^T(\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0}$

- **Viewpoint:** Calculus
  Key condition: Gradient of $\|A\mathbf{x} - \mathbf{b}\|^2$ is zero
  Equation: $A^T A\hat{\mathbf{x}} = A^T\mathbf{b}$

- **Viewpoint:** Hessian

Key condition: $2A^T A$ is positive definite if columns are independent
Equation: unique global minimum